

span > a.SQnoC3ObvgnGjWt9OzD9Z > h2” for the top post at the same URL, “<https://www.reddit.com/r/todayilearned/>” over time, or the knowledge that there is a pattern to the URLs of consecutive xkcd posts.

- **Step 2: Tool selection**

This is where the scripting starts... in a bit. A second type of recon is necessary. Just like a soldier needs to know their weapons and ammunition, you need to know your modules. (Or just look them up on the job.) When building functionality with code, the right tools sure make the job easier. You can directly use lxml or configure BeautifulSoup to use lxml as a parser for optimal performance and ease. (<https://www.crummy.com/software/BeautifulSoup/bs4/doc/#specifying-the-parser-to-use>)

If you’re not too concerned about performance (say you aren’t scraping around a million links), this could be a good chance to practice your use of regular expressions, instead of going with bs4. The re module is a dull ache to learn, but the rewards of doing so are aplenty! There are also timer modules that you can make a habit of inserting in your scripts to benchmark performance.

- **Step 3: Start Building**

Now the scripting really starts. Import the modules that you have chosen as your tools and start building your code one function at a time [most likely: requests, beautifulsoup/re]. Unlike many other languages, you don’t need a main() function, so you can start statement by statement and test functionality, before having to organize it into functions according to the control flow. First get() the data for a test page URL and store the HTML response in a variable. At the end print out a part of the page (like the heading) using your method of choice (re or bs4). Alternately, if you are fetching other types of files, such as music or images, you may need to use the os module to write the response to a file, and later to properly name them. The os module is very powerful so use it with care. We aren’t liable if you write a script to iteratively delete your system and can’t stop the recursion in time. That said, it will play a major role in the next project, so don’t be afraid to get familiar with it.

Once you have the basic single page request working to your will, change the action statement from printing out the heading to whatever you need to be done with the data. (Extract text/images, save a file, etc). At this point, start building a loop around the code. Of course, leave out the imports (and